Programming foundations practice guide

Original Kody exercises for tracing instructions, decisions, text and stacks.

Instructions and state

A program changes state by following instructions in order. In a game, state may include a position, a stored number, a list or a stack. Before running code, write down the starting state and predict the effect of each instruction.

A worked example

Start with score = 3. Add 2 to get 5. Multiply by 4 to get 20. Reversing the two instructions gives 14 instead. The same operations can produce different results when their order changes.

Conditions and Boolean gates

A condition chooses whether an action runs. AND is true when every input is true. OR is true when any input is true. XOR is true when an odd number of inputs is true. Test the cases where no inputs, one input and every input are on.

Text and stacks

Strings are ordered characters. Trimming removes spaces at the ends; case changes affect letters; reversing changes character order. A stack uses last in, first out. Push adds at the top and pop removes the newest value. In Memory Trail the rightmost displayed value is the top.

Practice tasks

1. Starting with 4, add 3 and multiply by 2. What is the result?
2. For inputs true, false, true, predict AND, OR and XOR.
3. Start with [1, 2], push 3, then pop. What remains?
4. Trim the text " code ", convert to uppercase and reverse it.

Explain your answer

The answers are 14; false, true, false; [1, 2]; and EDOC. Explain each transition rather than memorizing the final value. In Kody, building an instruction list is a preview. Press Run to validate an eligible completion. Reading a lesson advances course progress only after Mark as read; it does not award streak or XP credit.